

Hess's Law, Gibbs Free Energy, and Equilibrium Thermodynamics

Converted from Handwritten PDF

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1 Physical Chemistry – Study Sheet

1.1 Hess's Law

$$\Delta H_{rxn}^{\circ} = \sum \Delta H_f^{\circ} (\text{products}) - \sum \Delta H_f^{\circ} (\text{reactants})$$

- ΔH_f° of any element in standard state = 0
- Flip reaction $\rightarrow$ flip sign of ΔH°
- Scale reaction by $n \rightarrow$ scale ΔH° by n

1.2 Standard Entropy Change

$$\Delta S_{rxn}^{\circ} = \sum S^{\circ} (\text{products}) - \sum S^{\circ} (\text{reactants})$$

- Units: $\text{J mol}^{-1}\text{K}^{-1} \leftarrow$ watch the J vs kJ mismatch!
- More moles of gas $\Rightarrow \Delta S^{\circ} > 0$ (usually)

1.3 Gibbs Free Energy

Standard Conditions (298 K):

$$\Delta G^{\circ} = \Delta H^{\circ} - T\Delta S^{\circ}$$

Non-Standard Conditions:

$$\Delta G = \Delta G^{\circ} + RT \ln Q$$

$Q \equiv$ reaction quotient

ΔG	Meaning
< 0	spontaneous

ΔG	Meaning
$= 0$	equilibrium
> 0	non-spontaneous

2 Equilibrium Constant

$$\Delta G^\circ = -RT \ln K$$

$$\therefore K = e^{-\Delta G^\circ/RT}$$

- $R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$
- T must be in Kelvin
- $K > 1$: products favored; $K < 1$: reactants favored

3 Van't Hoff Equation

$$\ln \frac{K_2}{K_1} = -\frac{\Delta H^\circ}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$$

3.1 La Chatelier Shortcut:

Reaction Type	Raise T	Lower T
Exothermic ($\Delta H^\circ < 0$)	K decreases	K increases
Endothermic ($\Delta H^\circ > 0$)	K increases	K decreases

4 Unit Checklist

Symbol	Units
ΔH°	kJ mol^{-1}
ΔS°	$\text{J mol}^{-1} \text{K}^{-1}$ (convert to kJ before combining with ΔH°)
ΔG°	kJ mol^{-1}
R (with ΔG°)	$8.314 \times 10^{-3} \text{ kJ mol}^{-1} \text{K}^{-1}$ or convert ΔG° to J
T	Kelvin

Workflow

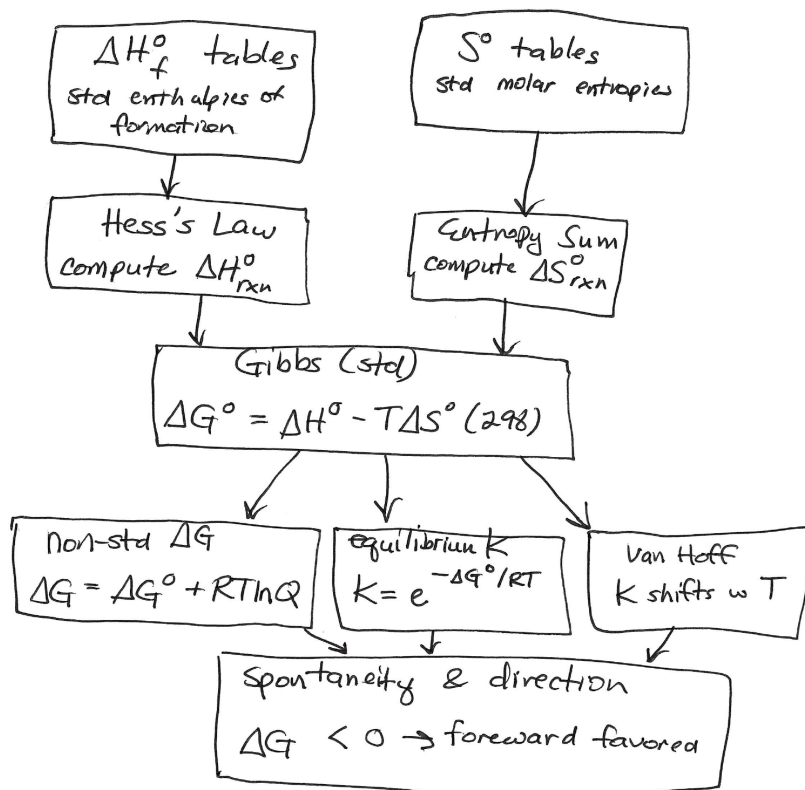


Figure 1: Diagram or drawing from handwritten notes

5 Workflow

{{FIGURE: Flowchart diagram showing the thermodynamic workflow described below}}

Top Level Inputs:

 ΔH_f° tables std enthalpies of
formation S° tables std molar entropies

Second Level (Computations):

- **Hess's Law** — compute ΔH_{rxn}°
 - **Entropy Sum** — compute ΔS_{rxn}°
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Third Level (Gibbs std):

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ \quad (298)$$

Fourth Level (Three Branches):**Left branch:** Non-std ΔG

$$\Delta G = \Delta G^\circ + RT \ln Q$$

Middle branch: Equilibrium K

$$K = e^{-\Delta G^\circ/RT}$$

Right branch: Van Hoff K shifts w T

Bottom Level:**Spontaneity & direction**

$$\Delta G < 0 \rightarrow \text{forward favored}$$